

AETHERIX

Autonomous Extraterrestrial High-throughput Enhancing Routing
and Inter-planetary eXchange

Building an Interplanetary Communication Network with DTN,
Quantum Communication, and Space-Based Infrastructure for Mars Mission Support

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matx104.github.io/AETHERIX | github.com/matx104/AETHERIX

PRESENTATION AGENDA

01 The Challenge

Why interplanetary communication is hard

02 AETHERIX Architecture

DTN + AI Routing + Quantum Security

03 DTN & Bundle Protocol v7

Store-and-forward networking foundation

04 5-Tier Network Topology

241 nodes across Earth and Mars

05 Optical Link Budget

1550 nm laser performance analysis

06 RL-Based Routing

Multi-agent federated Q-learning

07 Quantum Security (QKD)

BB84/E91 with repeater chains

08 Orbital Mechanics

Contact windows and synodic period

09 Mars Mission Scenario

End-to-end simulation walkthrough

10 Performance Comparison

AETHERIX vs current systems

11 Roadmap & Standards

CCSDS, IETF, deployment phases

12 Conclusion & Q&A

Summary and live demo

WHAT IS AETHERIX?

Autonomous Extraterrestrial High-throughput Enhancing Routing and Inter-planetary eXchange

What is AETHERIX?

An architecture for delay-tolerant networking (DTN) between Earth and Mars, combining:

- Bundle Protocol v7 (BPv7) via ION-DTN
- Reinforcement Learning routing (replacing static CGR)
- Quantum Key Distribution (QKD) for security
- Hybrid optical/RF communications
- 5-tier topology with 241 nodes

The Problem

- Earth-Mars: 54.6M – 401M km distance
- One-way light delay: 3 – 22 minutes
- Solar conjunction: 2-week total blackout
- TCP/IP fundamentally cannot work
- Current systems: 0.5 – 6 Mbps only

AETHERIX delivers a complete interplanetary networking solution

combining DTN protocols, AI-driven routing, quantum-secure keys, and hybrid optical/RF links for Mars mission support.

10-100×

Faster

>95%

Availability

241

Nodes

189

Tests

THE DISTANCE

Why Space Communication is Fundamentally Different

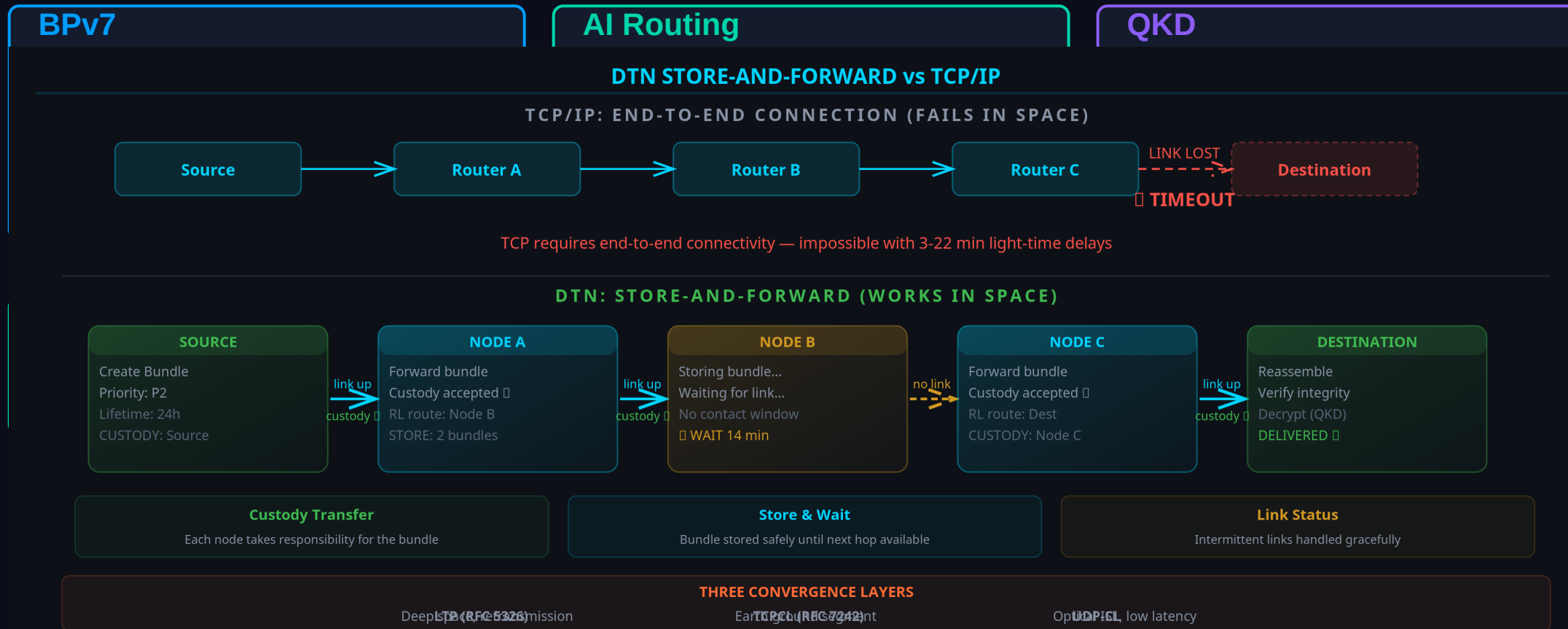
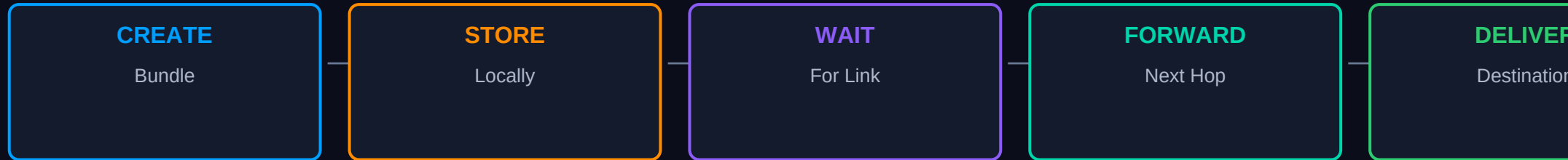
Earth → Mars	Distance	Light Time	Data Rate
Perihelion (closest)	54.6M km	~3 min	100-200 Mbps
Average	225M km	~12.5 min	10-20 Mbps
Aphelion (farthest)	401M km	~22 min	2-5 Mbps

TCP/IP Assumption	Space Reality	Impact
RTT < 1 second	6 – 44 min RTT	Timeout failure
Always connected	Scheduled contacts	Connection drops
End-to-end path	No persistent path	Routing impossible
Fast acknowledgments	ACKs take minutes	Window collapse
Low packet loss	High loss (optical)	Retransmit storms

"At aphelion, a single ping-pong takes 44 minutes. No TCP session can survive that. We need an entirely different networking paradigm."

THE ANSWER: DELAY-TOLERANT NETWORKING

Decouple Communication from Connectivity



SYSTEM ARCHITECTURE

5 Integrated Modules for Interplanetary Communication

Infrastructure

Link budget calculations, optical/RF analysis

Routing

RL agents, BPv7 bundles, forwarding engine

Security

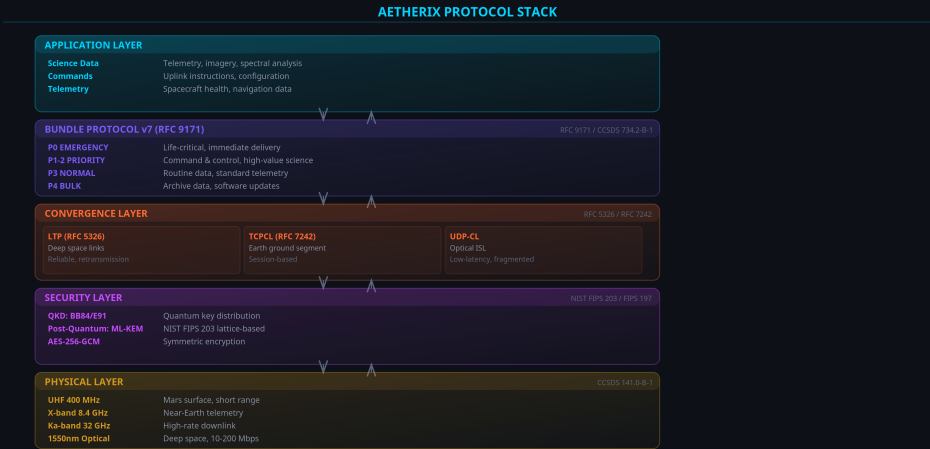
BB84/E91 QKD, repeater chains, privacy amplification

Orbital

Contact windows, Doppler, celestial body database

Simulation

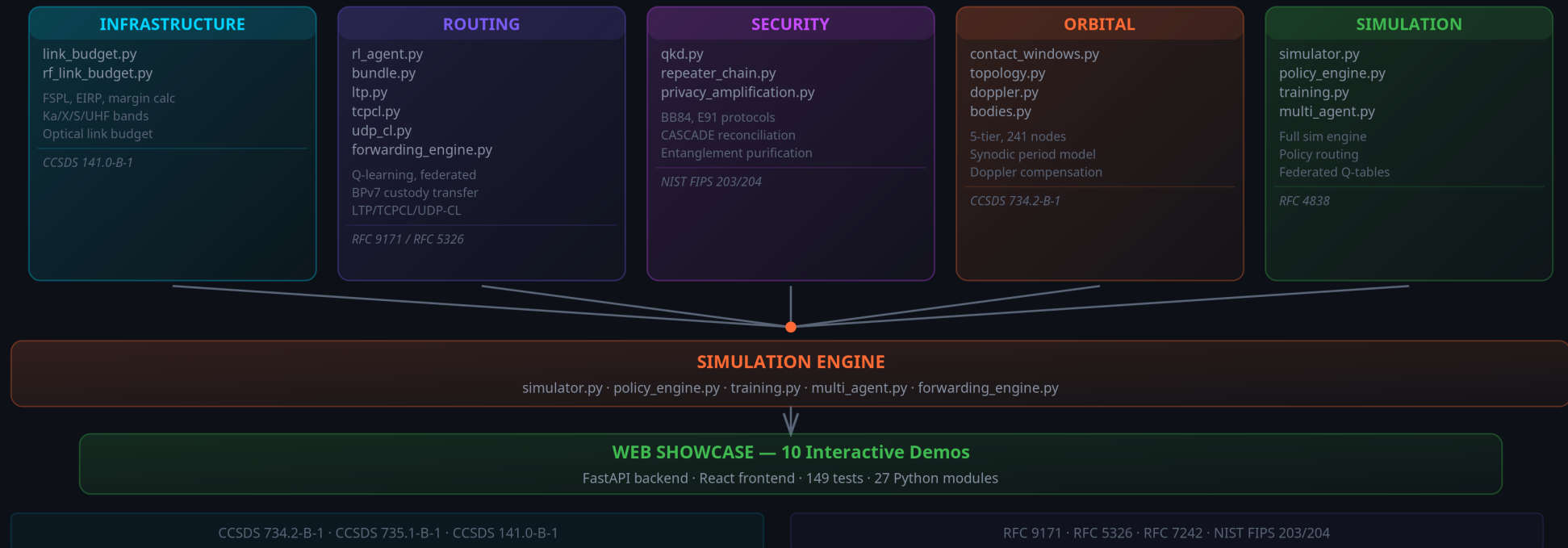
End-to-end scenario engine, policy routing



Layer	Module	Engine	Showcase
Infrastructure	link_budget.py	OpticalLinkBudget	Earth-Mars scenarios
Routing	rl_agent.py	RLRoutingAgent	Federated Q-learning
Security	qkd.py	BB84/E91	Quantum key exchange
Orbital	contact_windows.py	ContactPredictor	Synodic period
Simulation	simulator.py	SimEngine	Full mission sim

SYSTEM ARCHITECTURE DIAGRAM

AETHERIX SYSTEM ARCHITECTURE



BUNDLE PROTOCOL v7 — DEEP DIVE

RFC 9171 — The Foundation of Interplanetary Networking

APPLICATION LAYER

Science data, commands, imagery

BUNDLE PROTOCOL v7 (RFC 9171)

Store-and-Forward | Custody Transfer | Priority (P0–P4)

CONVERGENCE LAYERS

LTP (deep space) | TCPCL (Earth) | UDP-CL (optical ISL)

PHYSICAL LAYER

Optical 1550nm (2–200 Mbps) | RF Ka-band | UHF (surface)

HOW IT WORKS

1. Source creates bundle (payload + metadata + lifetime)
2. Forward to next hop when link becomes available
3. Store locally during outages and disruptions
4. Custody transfer — next node accepts responsibility
5. Repeat hop-by-hop until destination reached

STANDARDS COMPLIANCE

- RFC 9171 — Bundle Protocol Version 7
- RFC 4838 — DTN Architecture
- RFC 5326 — Licklider Transmission Protocol
- RFC 9172 — Bundle Protocol Security (BPSec)
- CCSDS 734.2-B-1 — CCSDS Bundle Protocol
- CCSDS 734.3-B-1 — Schedule-Aware Bundle Routing

Priority	Name	Delay Target	Example
P0	Emergency	< 1 min	Anomaly alerts
P1	Expedited	< 30 min	Solar storm warnings
P2	Standard	< 24 hrs	Science data
P3	Normal	< 7 days	Telemetry
P4	Bulk	< 30 days	Software updates

5-TIER NETWORK TOPOLOGY

241 Nodes — No Single Point of Failure

TIER 1: Earth Ground (6 nodes)

DSN Goldstone, Madrid, Canberra + Mission Control

TIER 2: Earth Orbital (51 nodes)

3 GEO relays + 48 LEO laser constellation

TIER 3: Deep Space (4 nodes)

ES-L4, ES-L5 Lagrange relays + Transfer orbit

TIER 4: Mars Orbital (4 nodes)

2 Areostationary + 2 polar orbiters

TIER 5: Mars Surface (176 nodes)

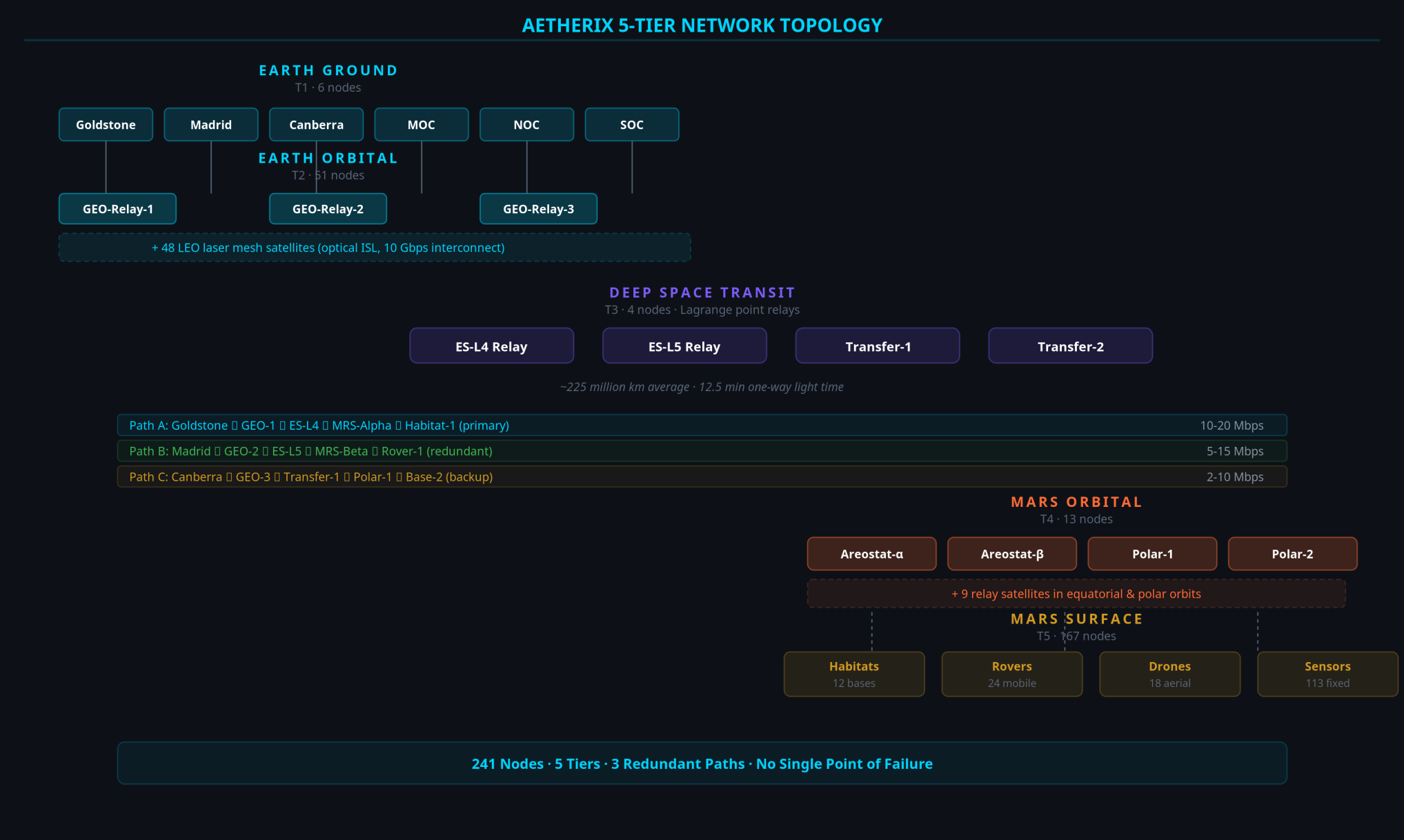
Bases, rovers, drones, sensor network

DESIGN PHILOSOPHY

Multiple redundant paths at every tier ensure no single point of failure. Lagrange point relays provide coverage during solar conjunction blackouts. Quantum repeaters co-located at L4/L5 enable end-to-end key distribution.

5-TIER NETWORK DIAGRAM

241 Nodes from Earth to Mars Surface



OPTICAL COMMUNICATIONS

1550 nm Laser — 10–100× Faster Than RF

200 Mbps

Peak Data Rate

At perihelion (54.6M km)

1550 nm

Wavelength

Telecom heritage lasers

33×

vs Current RF

MRO peak: 6 Mbps

KEY EQUATIONS

$FSPL = (4\pi d/\lambda)^2$ Free-Space Path Loss

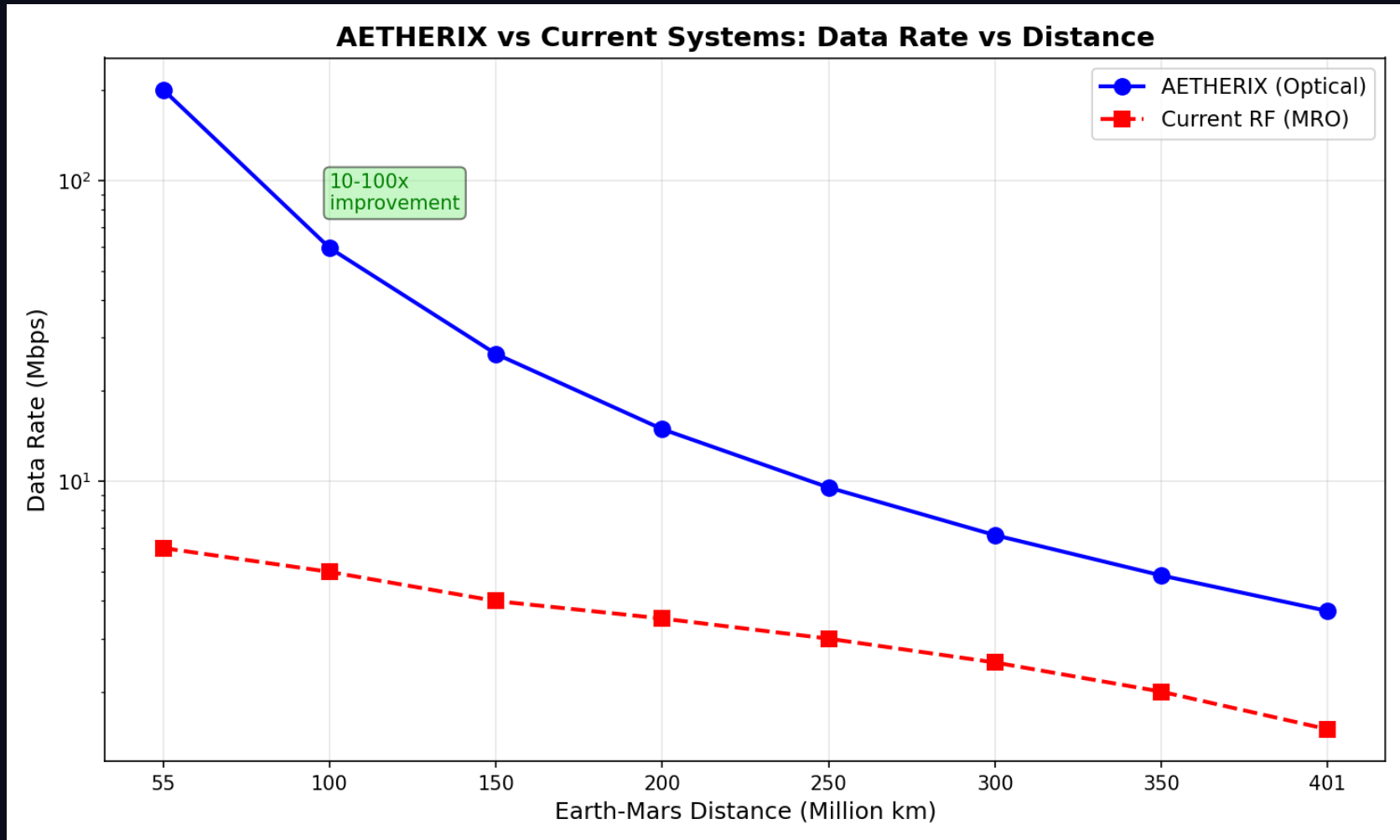
$G = (\pi D/\lambda)^2$ Antenna Gain

$P_r = P_t \cdot G_t \cdot G_r / FSPL$ Received Power

Parameter	Value	Rationale
TX Power	5 W (37 dBm)	Space-qualified laser
TX Aperture	22 cm	Mars orbiter class
RX Aperture	1.0 m	DSN-scale telescope
Wavelength	1550 nm	Telecom heritage
Efficiency	0.55	Conservative estimate

DATA RATE VS DISTANCE

1550 nm Optical Link Performance



200 Mbps at closest approach to 2 Mbps at maximum distance · 10–100× improvement over RF

EARTH-MARS DATA JOURNEY

7 Hops from Perseverance Rover to Mission Control

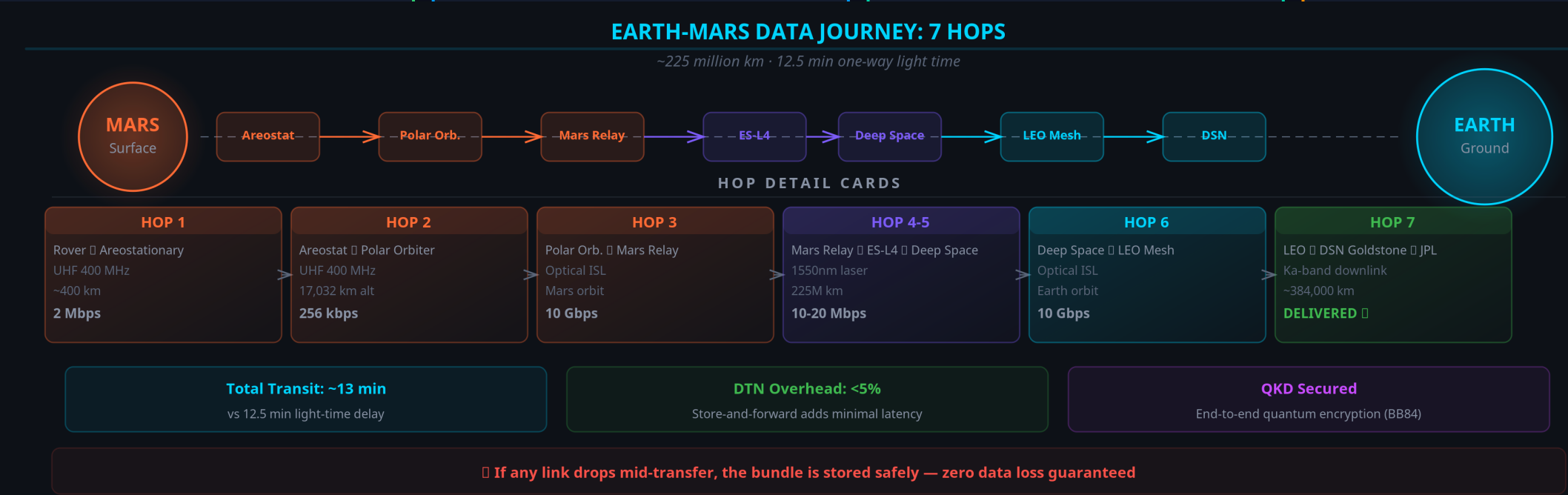
Hop	From	To	Link	Time
1	Perseverance	MRS-Alpha	UHF	~5s
2	MRS-Alpha	MRS-Polar	Optical ISL	~1s
3	MRS-Polar	ES-L5	1550nm laser	~12.5 min
4	ES-L5	ES-L4	Optical ISL	~2s
5	ES-L4	LEO Constellation	1550nm laser	~0.1s
6	LEO Constellation	DSN Madrid	Fiber downlink	~1s
7	DSN Madrid	JPL MOC	Terrestrial fiber	~0.5s

~13 min

98.7%

15 Mbps

7 hops



RL-BASED ROUTING — AI INNOVATION

Multi-Agent Federated Q-Learning Replaces Static CGR

CGR LIMITATIONS

- Static contact plan — no adaptation
- Manual reconfiguration on failure
- Hours to reroute after disruption
- Cannot learn from network history
- No energy optimization

RL AGENT ADVANTAGES

- Adaptive routing from network state
- 4 actions: FORWARD, STORE, DROP, SPLIT
- Reward: $\alpha(\text{delivery}) - \beta(\text{delay}) - \gamma(\text{hops})$
- Federated learning across agents
- Auto-recovery in seconds, not hours

97%

Delivery Ratio

+5% vs CGR

-20%

Delivery Time

1.2× light-time

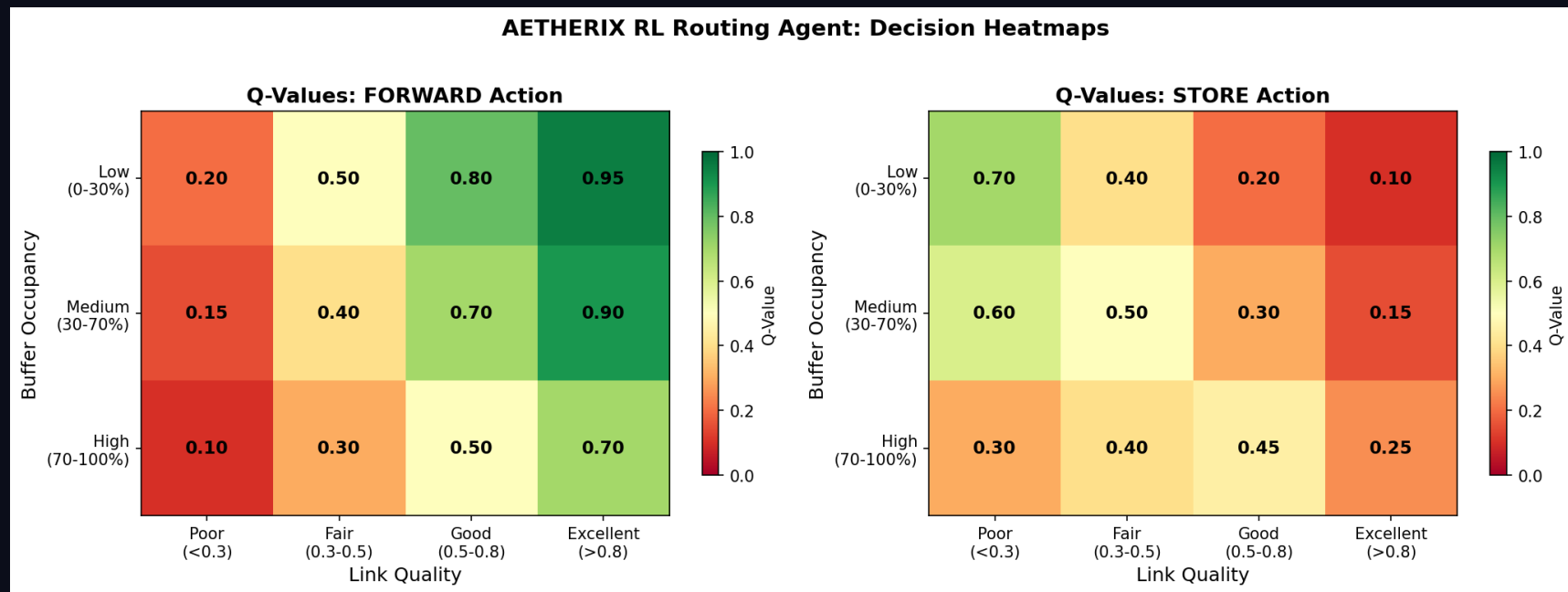
3600×

Recovery Speed

Auto vs manual

RL ROUTING Q-VALUE HEATMAP

Convergence of Optimal Routing Decisions



Warm colors = high-value actions · Cool colors = poor choices the agent avoids

QUANTUM SECURITY — FUTURE-PROOF

BB84/E91 QKD + Multi-Hop Repeaters + CASCADE

BB84 PROTOCOL STEPS

- 1. Alice sends qubits in random bases (rectilinear/diagonal)
- 2. Bob measures each qubit in random basis
- 3. Public reconciliation of basis choices (~50% retained)
- 4. QBER estimation: < 11% = SECURE, ≥ 11% = ABORT
- 5. CASCADE error reconciliation
- 6. Privacy amplification via universal hashing
- 7. Final shared secret key

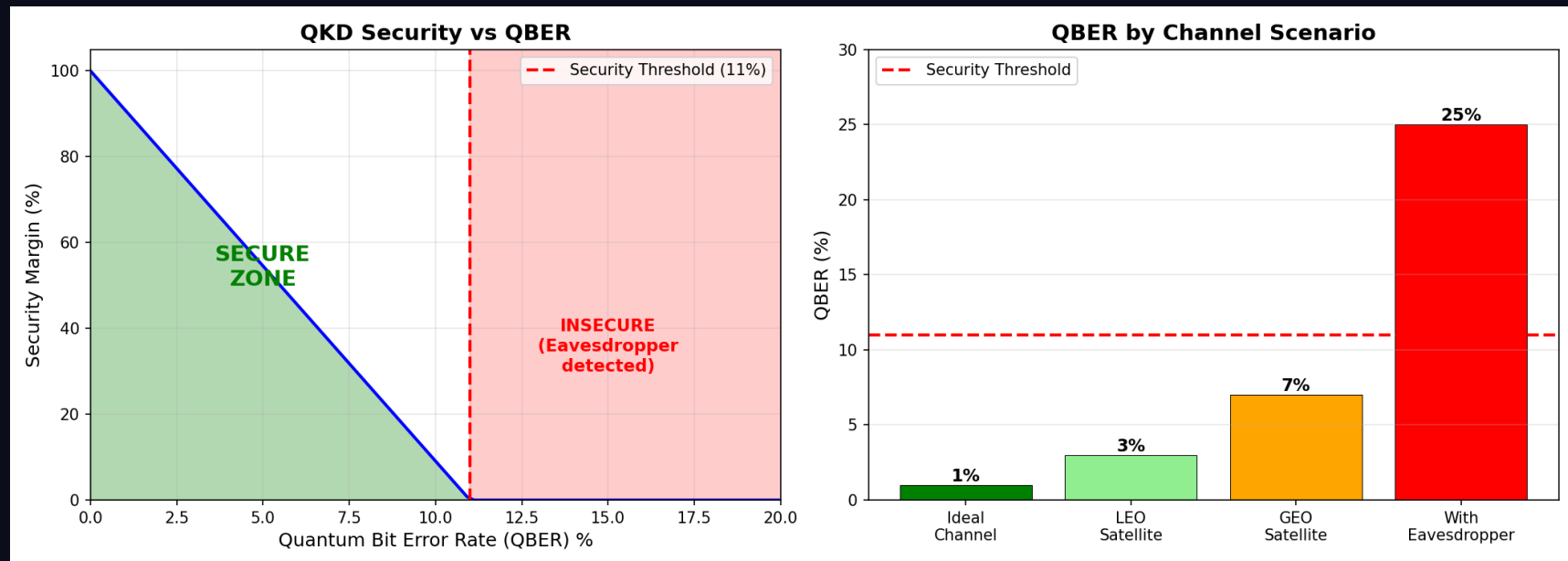
Phase	Segment	Protocol	Range
Phase 1	Earth ↔ Orbit	BB84 direct	36,000 km
Phase 2	Earth ↔ Lagrange	1 repeater	1.5M km
Phase 3	Earth ↔ Mars	3-hop chain	225M km

POST-QUANTUM CRYPTO (PQC)

Kyber (ML-KEM) — Key encapsulation
Dilithium (ML-DSA) — Digital signatures
Hybrid: QKD + PQC for defense in depth

QKD SECURITY ANALYSIS

QBER vs Eavesdropper Detection



QBER < 11% threshold ensures security · Any eavesdropping attempt is detected

ORBITAL MECHANICS & CONTACT WINDOWS

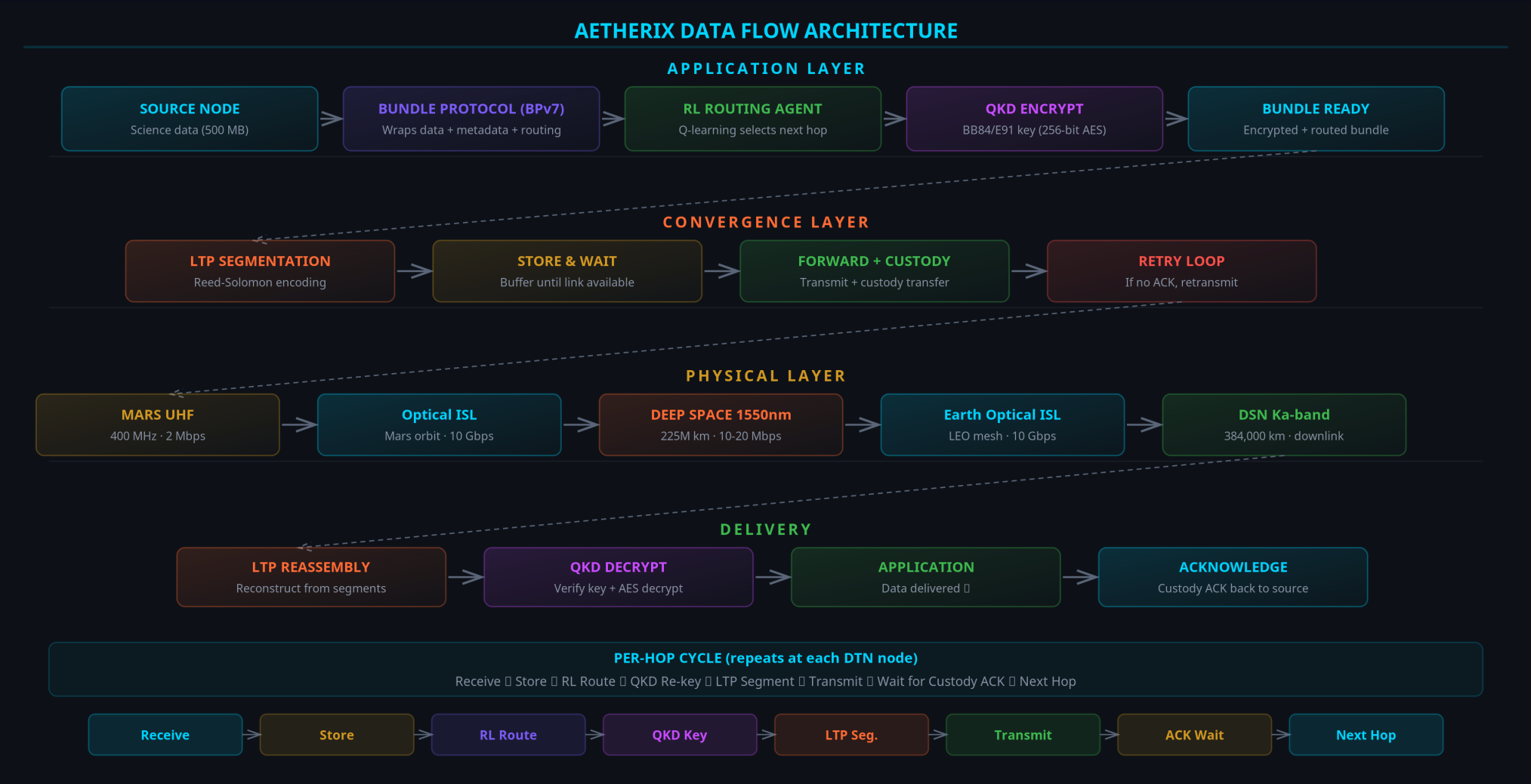
780-Day Synodic Period — From Opposition to Conjunction

Parameter	Value
Semi-major axis	1.524 AU (227.9M km)
Orbital period	686.98 days
Synodic period	779.94 days
Min distance	54.6M km (perihelion)
Max distance	401M km (aphelion)
Eccentricity	0.0934

Window	Availability	Duration	Data Rate
Optimal (opposition)	99%	8–12 hrs	100–200 Mbps
Good	95%	6–8 hrs	20–100 Mbps
Fair	85%	2–4 hrs	5–20 Mbps
Poor (aphelion)	50%	0–2 hrs	1–5 Mbps
Blackout (conjunction)	0%	N/A	Lagrange relay only

DATA FLOW DIAGRAM

Complete Data Path from Mars Surface to Earth Control



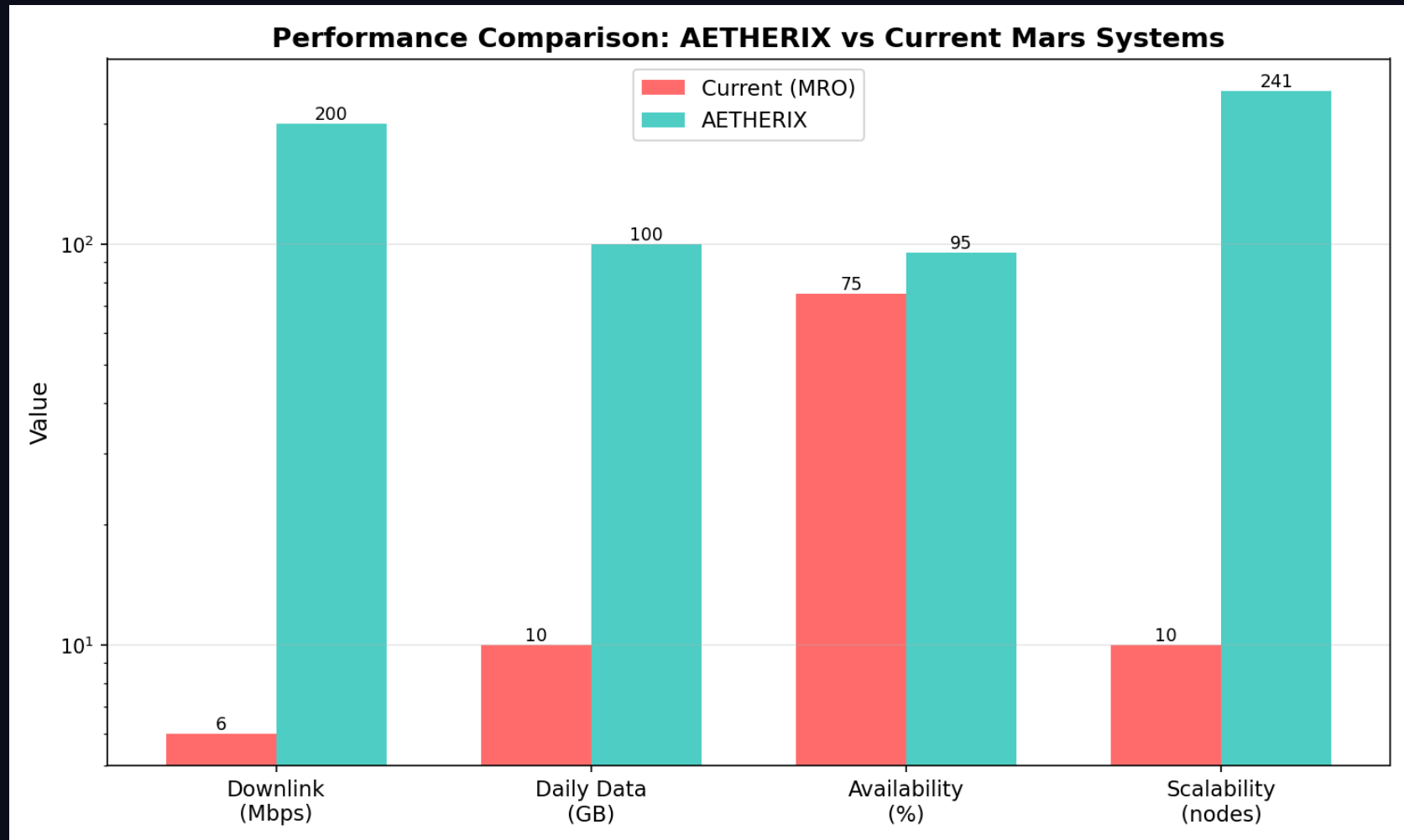
AETHERIX vs CURRENT SYSTEMS

Head-to-Head Performance Comparison

Metric	Current (MRO)	AETHERIX	Improvement
Downlink rate	0.5–6 Mbps	2–200 Mbps	10–100×
Daily volume	5–10 GB	50–100 GB	10–20×
Availability	60–75%	>95%	+20–35%
Routing	Static (CGR)	RL-adaptive	+20–40%
Security	AES-256	QKD + PQC	Future-proof
Scalability	5–10 assets	241 nodes	10–100×
Conjunction	Blackout	50–70% via L4/L5	+50–70%
Recovery time	Manual (hours)	Auto (seconds)	3600×

AETHERIX vs CURRENT SYSTEMS

Head-to-Head Performance Metrics



10–100× improvement across all metrics

IMPLEMENTATION

27 Modules, 189 Tests, Full Python Stack

27

Source Modules

189

Unit Tests

241

Network Nodes

12

Interactive Demos

CORE MODULES

- link_budget.py — Optical/RF link analysis
- rl_agent.py — Q-learning routing agent
- bundle.py — BPv7 bundle protocol
- forwarding_engine.py — Store-and-forward
- qkd.py — BB84/E91 protocols
- contact_windows.py — Orbital mechanics
- topology.py — 5-tier network (241 nodes)
- simulator.py — End-to-end simulation
- ltp.py — Licklider Transmission Protocol
- tcpcl.py — TCP Convergence Layer

Standard	Title	Status
RFC 9171	Bundle Protocol v7	Compliant
RFC 4838	DTN Architecture	Compliant
CCSDS 734.2-B-1	CCSDS Bundle Protocol	Compliant
CCSDS 734.3-B-1	SABR / contact routing	Baseline
CCSDS 141.0-B-1	Optical Comm Physical	Compliant
RFC 9172	BPsec (security)	Referenced
RFC 5326	LTP (deep space)	Compliant
RFC 7242	TCPCL (Earth)	Compliant

ROADMAP & FUTURE WORK

From Proof-of-Concept to Production Deployment

Phase 1–2: Foundation
Topology + BPv7 + Convergence Layers

Complete

Phase 3: Intelligence
RL Routing (Federated Q-learning)

Complete

Phase 4: Security
QKD + Multi-hop Repeaters + CASCADE

Complete

Phase 5–6: Simulation
Full Engine + Web Platform + Demos

Complete

Phase	Focus	Technology	Status
7	DQN Upgrade	Deep Q-Network	Remaining
8	HIL Simulation	ns-3 integration	Remaining
9	ION-DTN	Production DTN stack	Remaining
10	Flight Demo	LEO optical test	Future
11	Mars Deploy	Full 5-tier network	Future

CONCLUSION — AETHERIX DELIVERS

Bridging the Interplanetary Communication Gap

THE PROBLEM

- 54.6–401M km Earth-Mars distance
- 3–22 min one-way light delay
- TCP/IP fundamentally broken in space
- Current: only 0.5–6 Mbps
- 2-week solar conjunction blackouts

THE SOLUTION

- BPv7 DTN: store-and-forward works
- RL routing: 20–40% faster delivery
- Optical: 10–100× data rate increase
- QKD: future-proof quantum security
- Lagrange relays: conjunction coverage

10–100×

Faster Rates

>95%

Availability

AI-Driven

Routing

Quantum

Security

NOVEL CONTRIBUTIONS

RL autonomous routing (20-40% faster) | 5-tier DTN (>95% availability) | Optical/RF hybrid (10-100x) | Quantum links (future-proof) | Full simulation (189 tests)

THANK YOU

Questions?

10–100×

Faster

>95%

Available

241

Nodes

189

Tests

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LIVE DEMO & RESOURCES

matx104.github.io/AETHERIX — Interactive simulations

github.com/matx104/AETHERIX — Source code + documentation

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